Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8004
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A & B)Mech / Civil / CSE / AI &ML / EEE (NEP) Sem - I&II
Examination June/July 2025
Engineering Chemistry

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q. nos 2, 3, 4, 5, 6and 7

SECTION A

Q1 Solve the following questions **20**

a) Which of the following is an addition polymer?
 a) Polyvinyl chloride (PVC) b) Bakelite
 c) Phenol formaldehyde d) nylon

b) Which of the following is not proximate analysis of coal?
 a) %ash b) %moisture
 c)% volatile matter d) % Sulfur

c) ________is the process where heating of metals takes place in the absence of air.
d) Write two properties of PVC and PVa.
e) What is caustic embrittlement?
f) Define sintering.
g) What is coolant? Give example of coolant.
h) What do you mean by pH.? Draw pH scale.
i) What is refining of metals? Enlist two methods of refining of metals.
j) Draw the structure of M-EDTA complex
k) Write any two properties and applications of CNG.
l) Define dry corrosion and wet corrosion.

SECTION B

Q2 a) Discuss the methods to prevent corrosion- powder coating **05**
b) Give the properties and applications of fiber glass and silicon carbide. **05**

Q3 a) Explain determination of % moisture and % ash as a part of proximate analysis of coal **05**
b) b) Explain the working mechanism of an alkaline fuel cell. **05**

Q4 a) Describe the working and applications of a lithium-ion battery. **05**
b) Differentiate between thermosetting and thermoplastic polymers. **05**

Q5 a) Calculate temporary, permanent and total hardness of water sample having following data - $MgSO_4$ = 130 ppm, $Ca(HCO_3)_2$= 120 ppm, $MgCl_2$ = 190 ppm, $CaCO_3$ = 130 ppm **05**
b) 50 ml of std hard water containing 1.5 mg of $CaCO_3$ per ml consumed 20 ml of EDTA. 50 ml of hard water sample consumed 15 ml of EDTA using EBT as indicator. 50 ml of boiled water sample consumed 10 ml of EDTA solution Calculate temporary, permanent and temporary hardness of water. **05**

Q6	a)	Explain the ion exchange process of water softening.	**05**
	b)	Write a note on ultrafiltration and its applications in water treatment.	**05**
Q7	a)	Explain the terms of scale, sludge, priming and foaming.	**05**
	b)	An oil of unknown viscosity index has saybolt universal viscosity of 60 Sec. at 2100F and 700 Sec. at 1000F. The high viscosity index standard (i.e. Pennsylvanian) oil has saybolt viscosity of 60 Sec. at 2100F and 600 Sec. at 1000F. The low viscosity index standard (i.e. Gulf) oil has a saybolt universal viscosity of 60 Sec. at 2100F and 900 Sec. at 1000F. Calculate the viscosity index of unknown oil.	**05**